

4 Motion in a circle

What students need to learn:

Angular speed.

Radial acceleration in circular motion. The forms $r\omega^2$

and $\frac{v^2}{r}$ are required.

Uniform motion of a particle moving in a horizontal circle.

Problems involving the 'conical pendulum', an elastic string, motion on a banked surface, as well as other contexts, may be set.

Motion of a particle in a vertical circle.

5 Statics of rigid bodies

What students need to learn:

Centre of mass of uniform rigid bodies and simple composite bodies.

The use of integration and/or symmetry to determine the centre of mass of a uniform body will be required.

Simple cases of equilibrium of rigid bodies.

To include

- (i) suspension of a body from a fixed point,
- (ii) a rigid body placed on a horizontal or inclined plane.

Unit M4 Mechanics 4

GCE AS and GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional) A2 optional unit

M4.1 Unit description

Relative motion; elastic collisions in two dimensions; further motion of particles in one dimension; stability.

M4.2 Assessment information

Prerequisites

A knowledge of the specifications for M1, M2 and M3 and their prerequisites and associated formulae, together with a knowledge of the calculus covered in FP2 and $\int \frac{1}{a^2 + x^2} dx$, is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. It will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: +, −, ×, ÷, π, x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

$${}_A\mathbf{v}_B = \mathbf{v}_A - \mathbf{v}_B$$